

Rahul Rawat

rahulrawat2r@gmail.com • 015563603340 • [linkedin.com/in/rahulrawat2r](https://www.linkedin.com/in/rahulrawat2r) • github.com/rahulrawat20022002 • Mannheim, Germany

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Leica Microsystems

Hiring Team

Application: Master Thesis Student, Data Science, Preventative Service

Dear Hiring Team,

I am applying for the Master Thesis Student position in Data Science focused on Preventative Service at Leica Microsystems in Mannheim. I already live in Mannheim, I am enrolled in the M.Sc. Data Science and Analytics programme at SRH Heidelberg, and the brief on preventive and predictive maintenance for microscope defects is exactly the kind of thesis topic I have been actively preparing for.

For the past six months I have been the Data Scientist on a collaboration between eRay GmbH and SRH University Heidelberg, building an end to end recursive forecasting pipeline for four water quality signals on real sensor data. I benchmarked six models including CatBoost with multi quantile regression, enforced strict anti leakage rules across the pipeline, reconstructed missing winter readings with MICE imputation, and wrapped the whole workflow in an orchestrator with gate checks that halt on failed imputation rather than letting bad data cascade downstream. That project translates directly into using device and service data to predict and prevent customer downtime.

My CreditIQ project shows the same discipline applied to decision support. I lifted a fairness metric from 0.79 to 0.88, cut the false negative rate from 44 percent to 16.7 percent while holding accuracy at 75 percent, and shipped a Streamlit tool that gives operators a recommendation together with a plain language explanation, keeping a human in the loop. I have also built a real time flight tracking pipeline on Google Cloud, joining device level positions to airport, aircraft, and weather data every 30 seconds through PySpark and dbt, which is close to how a service data warehouse gets fed. My Bachelor Thesis on diabetes prediction, written up as an IEEE style paper, proves I can drive a thesis from literature review to honest limitations section end to end.

I would be glad to visit the Mannheim site to discuss the thesis scope in person. Thank you for considering my application.

Kind regards,

Rahul Rawat